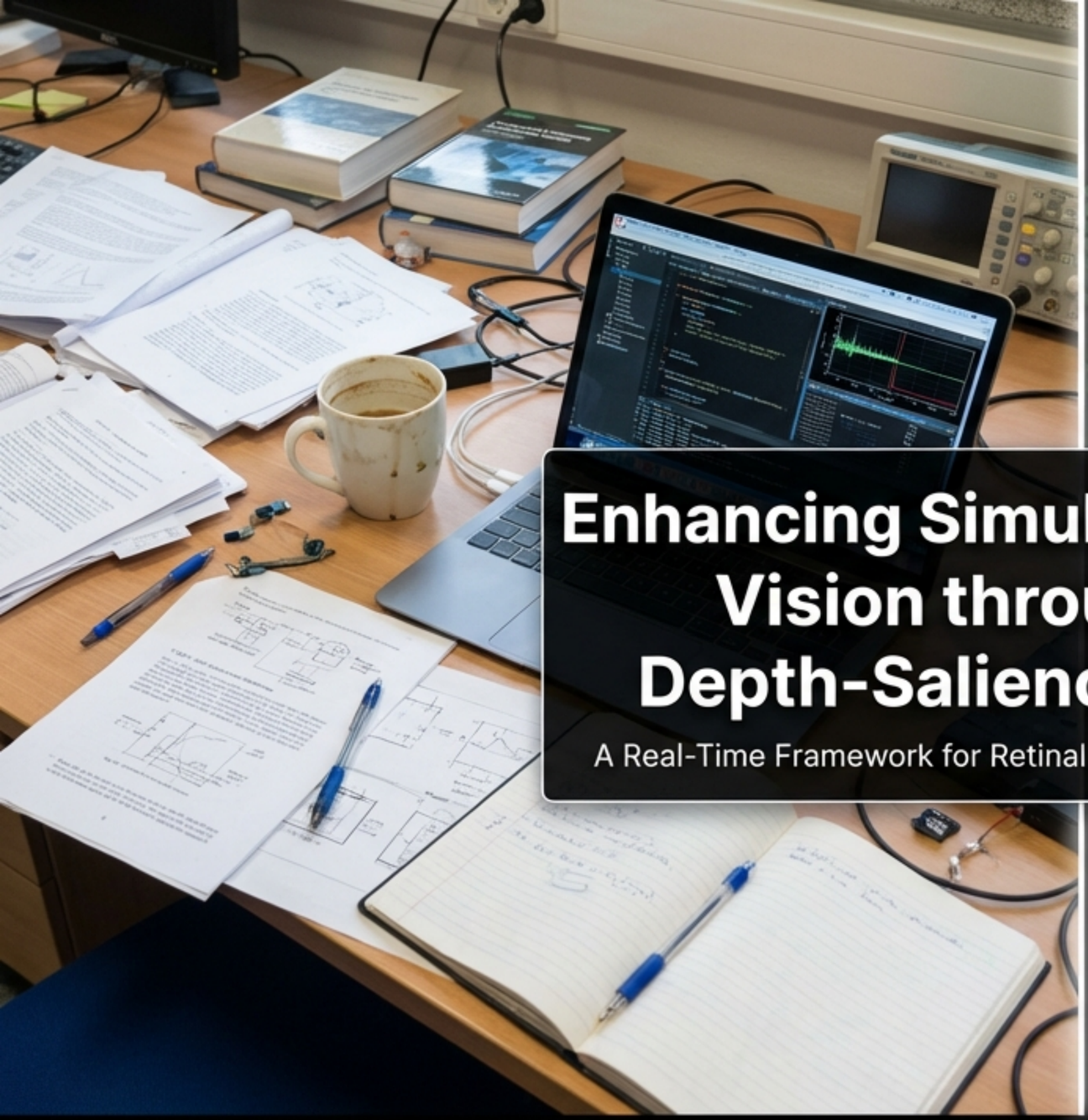


Enhancing Simulated Prosthetic Vision through Hybrid Depth-Saliency Processing

A Real-Time Framework for Retinal Implant Simulation and Validation

Group 2: Edoardo Capobianco, Simona Frascella, Michele Ivo Galella

Project Scope & Objective

- **Context:** Retinal prostheses (bionic eyes) offer hope for the blind but suffer from low resolution and noisy signals.
- **Problem:** Developing optimal encoding algorithms requires surgical patients, which is invasive and slow.
- **Solution - Simulated Prosthetic Vision (SPV):** A computational proxy that transforms video streams into phosphene patterns, mimicking the percepts of patients with epiretinal prostheses [Chen et al., 2009].

The Engineering Gap

Computer Vision
(Perfect Input)

The Biological Constraint

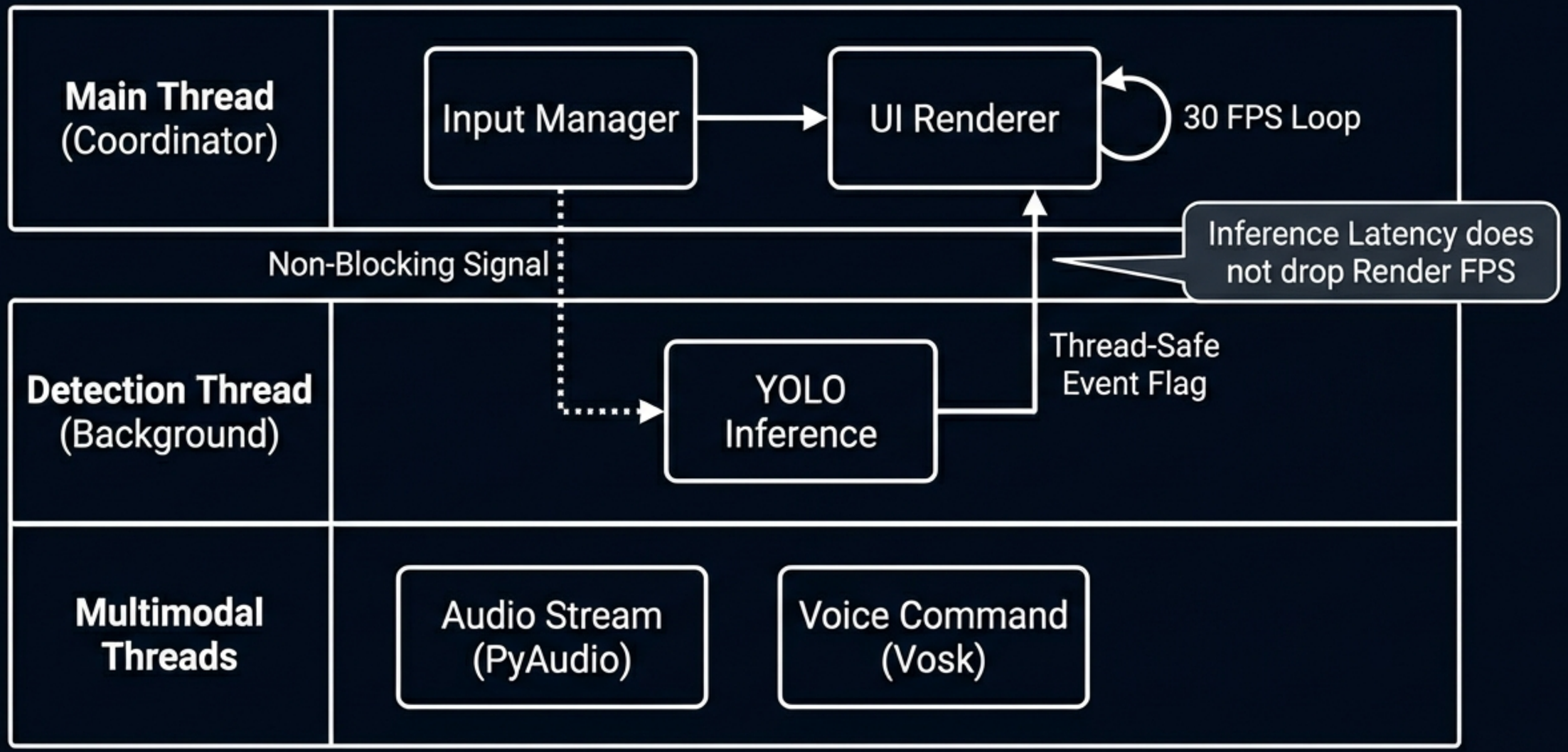
Neural Perception
(Limited Output)

Core Objectives

- Engineer a real-time (30 FPS) simulator.
- Integrate Semantic AI and Depth Sensing.
- Validate via biophysical rendering.
- Reduce visual clutter in low-res (32x32) grids.

Asynchronous System Architecture

Decoupled Inference for Real-Time Performance



Phase 1: Hybrid Acquisition

RGB + LiDAR Fusion (Calibration-Free)

RGB Texture
(Context)

+

Depth Map
(Metric Distance)

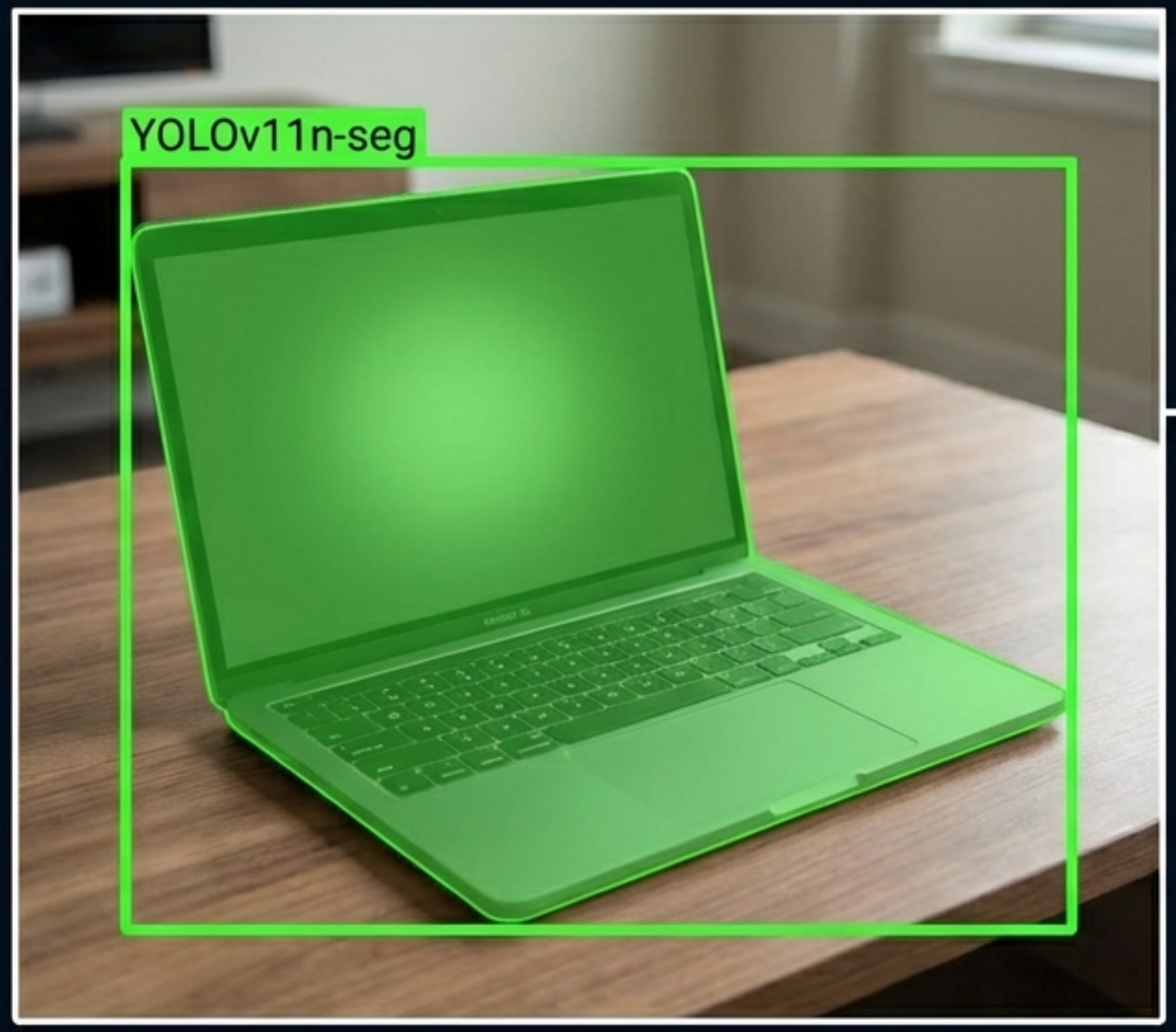
+

Confidence Map
(Reliability)

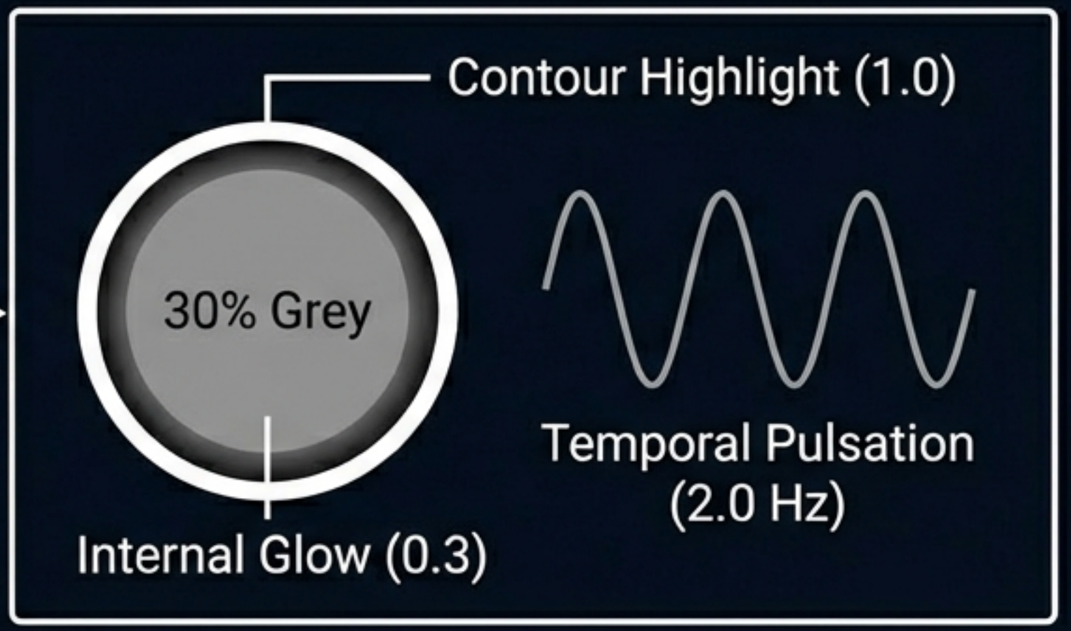
```
> Source: IPhoneLiDARStream (USB Interface)
> Fallback: MockLiDARStream (Debug only)
> Resolution: 640px (Resized)
> Aspect Ratio: 1:1 Square Crop
> Rotation: 90° (Mobile Landscape)
```

Phase 2: Semantic Perception

From Bottom-Up Saliency to Task-Driven Attention



Visual Encoding Strategy

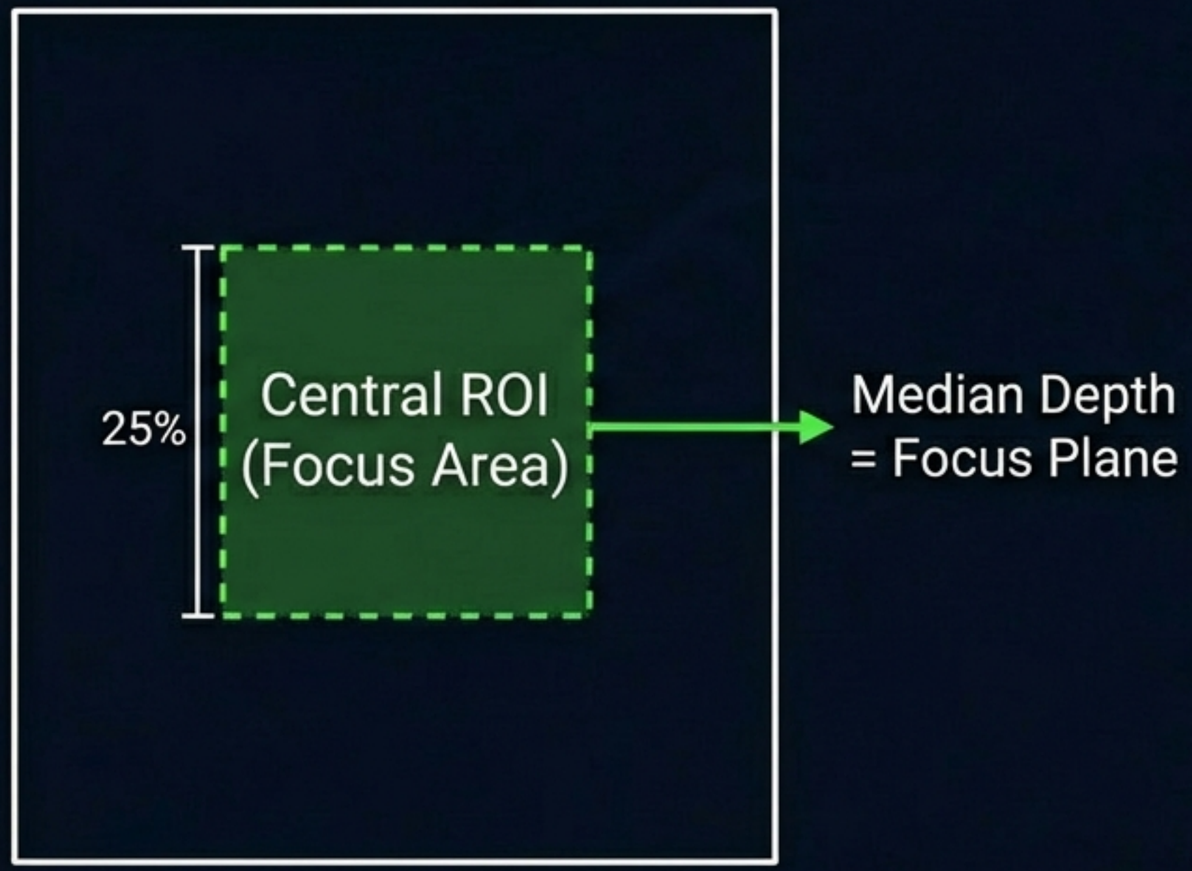


Why? The retina is highly sensitive to flicker. Pulsating the target effectively **segregates** it from the static background without needing color.

Phase 3: Depth Processing

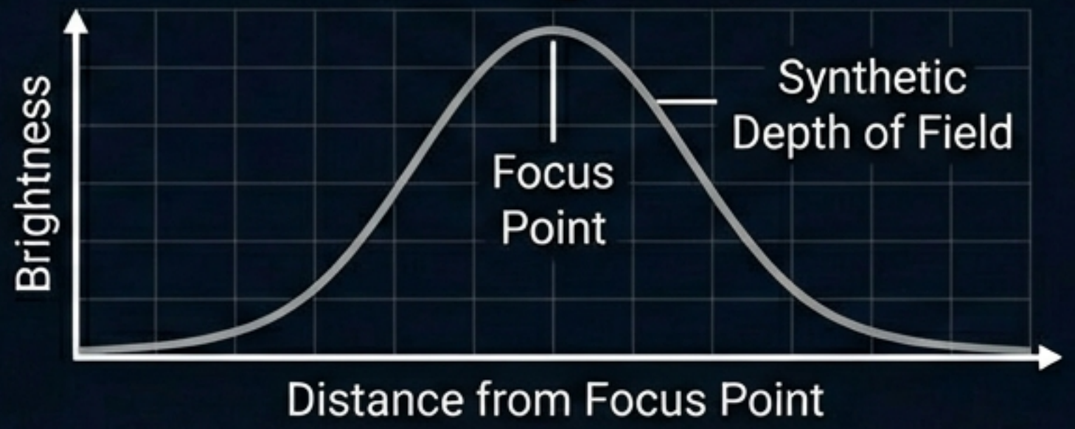
Dynamic Focus & Auto-Range

Scene Map



Inertial Smoothing (EMA)

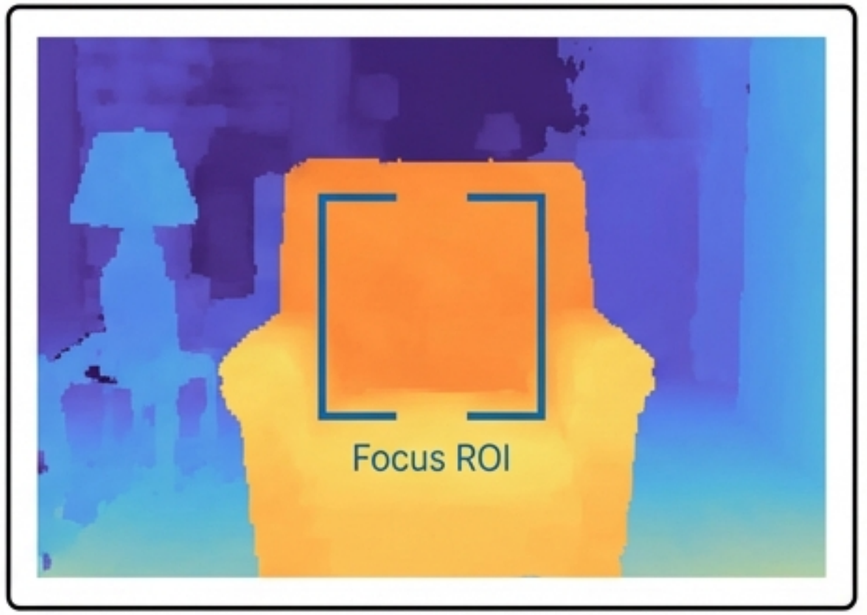
Dimming Curve

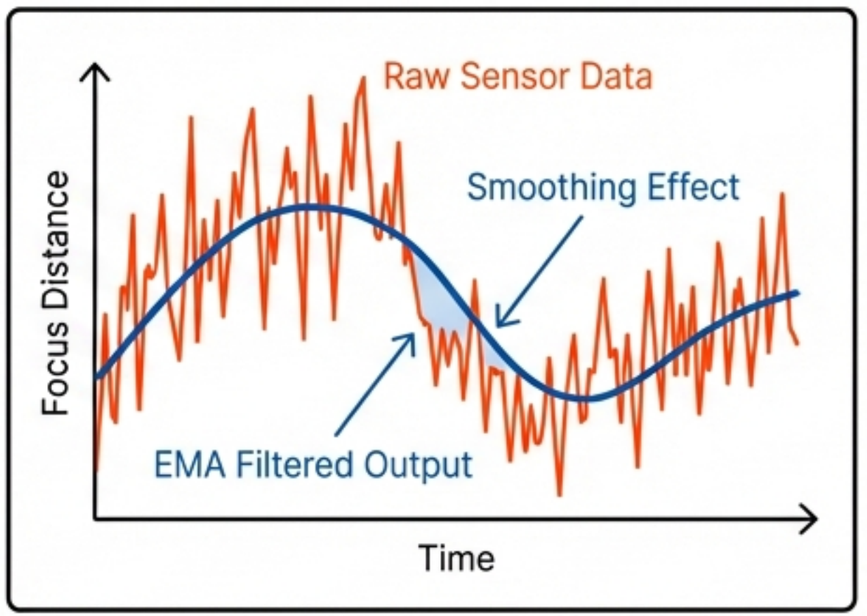


Dynamic Focus & Scene Understanding

The “Auto-Range” Algorithm:

- **ROI Analysis**: Calculates median depth in central 25% (`focus_roi_percentage`).
- **Exponential Moving Average (EMA)**:
  `smooth_focus_dist = 0.9 * smooth_prev + 0.1 * current`
- **Purpose**: Imparts inertia to focus point, preventing nausea-inducing jumps from sensor noise.





Synthetic Depth of Field:

Brightness modulated by focus distance.
Steep gradient for near-focus isolation.

Structural Edge Extraction

The Hybrid Engine: Structure over Texture

Mode A: Environmental (> 1.0m)

Algorithm: Asymmetric Depth-of-Field

$$Edge_{out} = \frac{Gradient}{(NoiseFloor + Penalty_{dof})}$$

Suppresses edges beyond the focal plane to reduce background clutter.

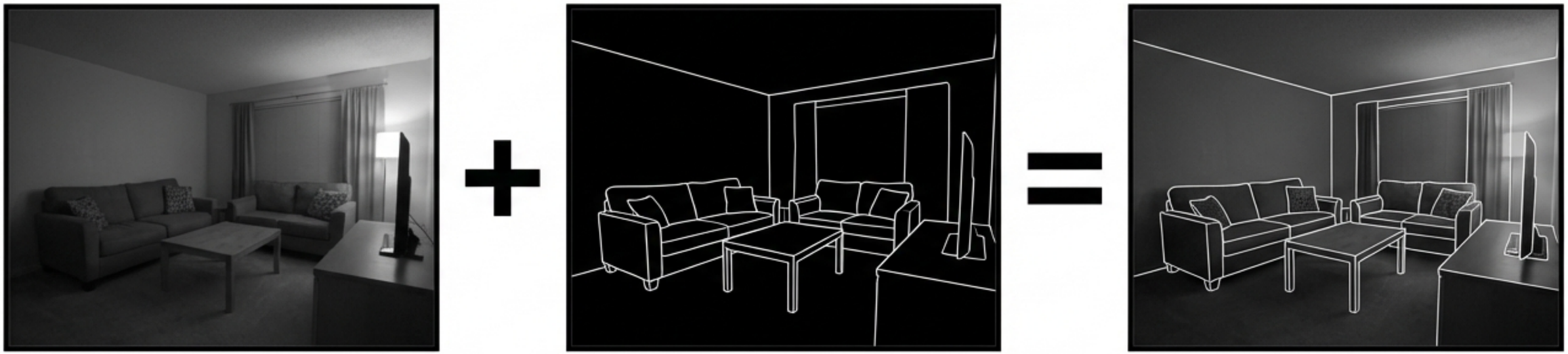
Mode B: Manipulation (< 1.2m)

Algorithm: Artifact Correction (FaceID/TrueDepth)

[Raw Depth with Black Halo Artifacts] → [Morphological Erosion] → [Bilateral Filter] → [Canny Edge Detector]

Fixes “flying pixels” and smooths surfaces while keeping edges sharp.

Hybrid Fusion Strategy



Texture (RGB) + Structure (Depth Edges) = Fused Scene Map

```
spv_input = np.maximum(Texture, Structure)
```

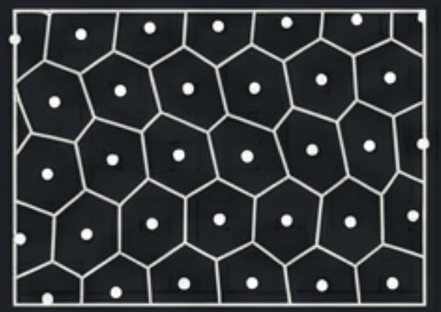
Engineering Justification: Prioritizes structural contours (Value 1.0) over surface textures. Ensures safety-critical edges (tables, walls) are never overwritten by dark surface colors.

Pipeline Stage 4: Biologically Realistic Rendering

Module: phosphene_processing_v4.py

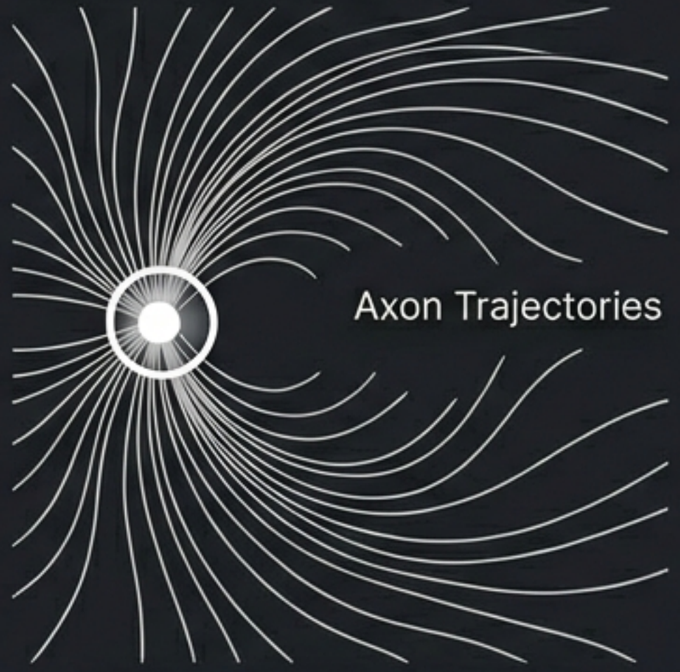
The Grid

Standard



Jittered Hexagonal - Biologically Plausible

The Axon Map




```
Angle = np.arctan2(dy, dx)
```

The Phosphene

```
sigma_y = current_sigma * elongation
```

Advanced Psychophysics

Blooming & Temporal Decay

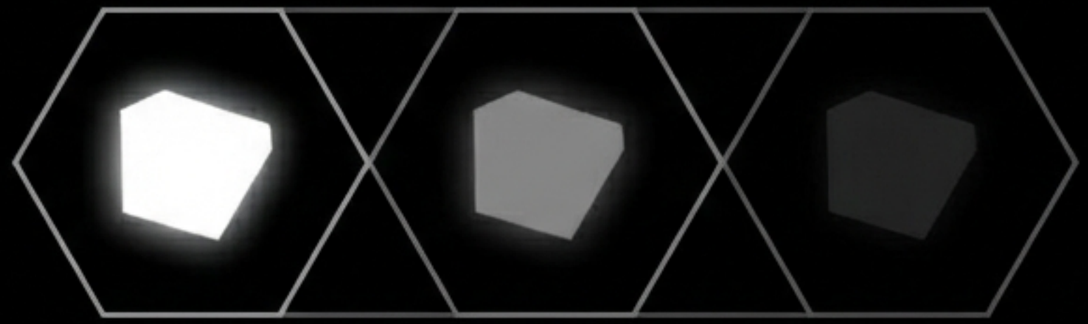
Luminance Blooming

Low Intensity

High Intensity

Size ~ 0.8 + 0.7 * Luminance

Temporal Decay (Adaptation)



Static images fade to encourage active scanning (saccades).

[Based on psychophysics principles]

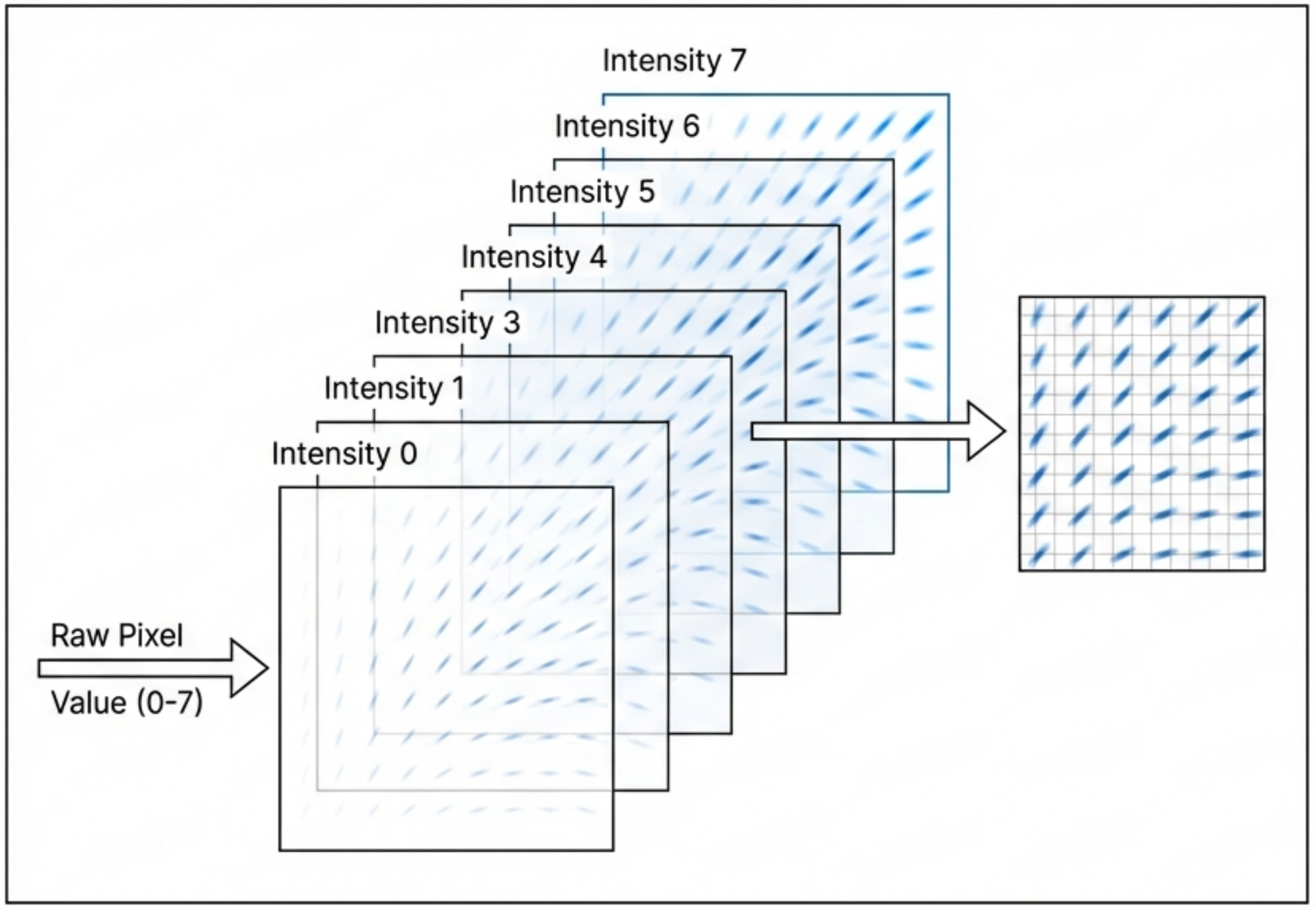
Real-Time Optimization (LUT)

Challenge: Calculating rotated Gaussians for 1000+ electrodes per frame is too slow for Python.

Solution: Look-Up Table (LUT) System.

1. Pre-computation: Kernels calculated for every electrode at 8 intensity levels during init.

2. Runtime Pipeline:
- Downsample (32x32).
- Normalize & Quantize.
- O(1) Retrieval from LUT.
- Composite via np.maximum.



Multimodal Interaction Layers

Audio & Voice Integration

Audio Feedback (Sonification)

Theremin-style proximity feedback.
Pitch and rate are inversely
proportional to distance.

Voice Control (Hands-Free)

```
> Wake Word:
  "Jarvis"
> Command:
  "Start / Stop"
> Command:
  "Cup / Bottle"
  (Target Selection)
```

Offline recognition using Vosk.
Essential for blind users.

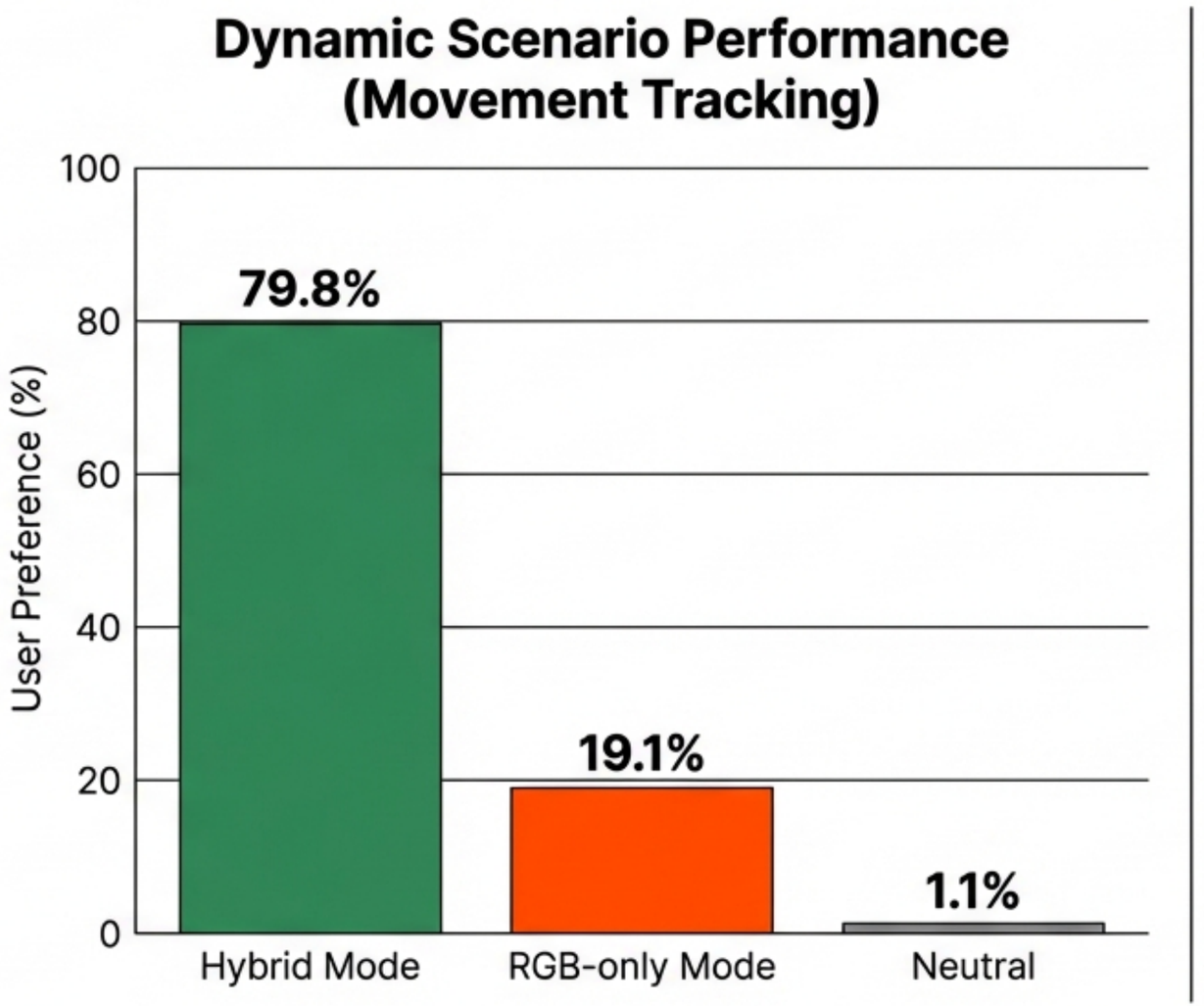
[Based on accessible interface guidelines]

Experimental Results: Dynamic Tracking

Context: 89 Participants. Comparison of RGB-only vs. Hybrid (Depth-Saliency) in motion.

Key Finding: 79.8% Preference for Hybrid Mode.

Analysis: Hybrid Mode suppresses 'ghosting' artifacts and temporal flicker. Users prioritized signal stability over texture detail in dynamic scenarios.



Experimental Results: Static Perception

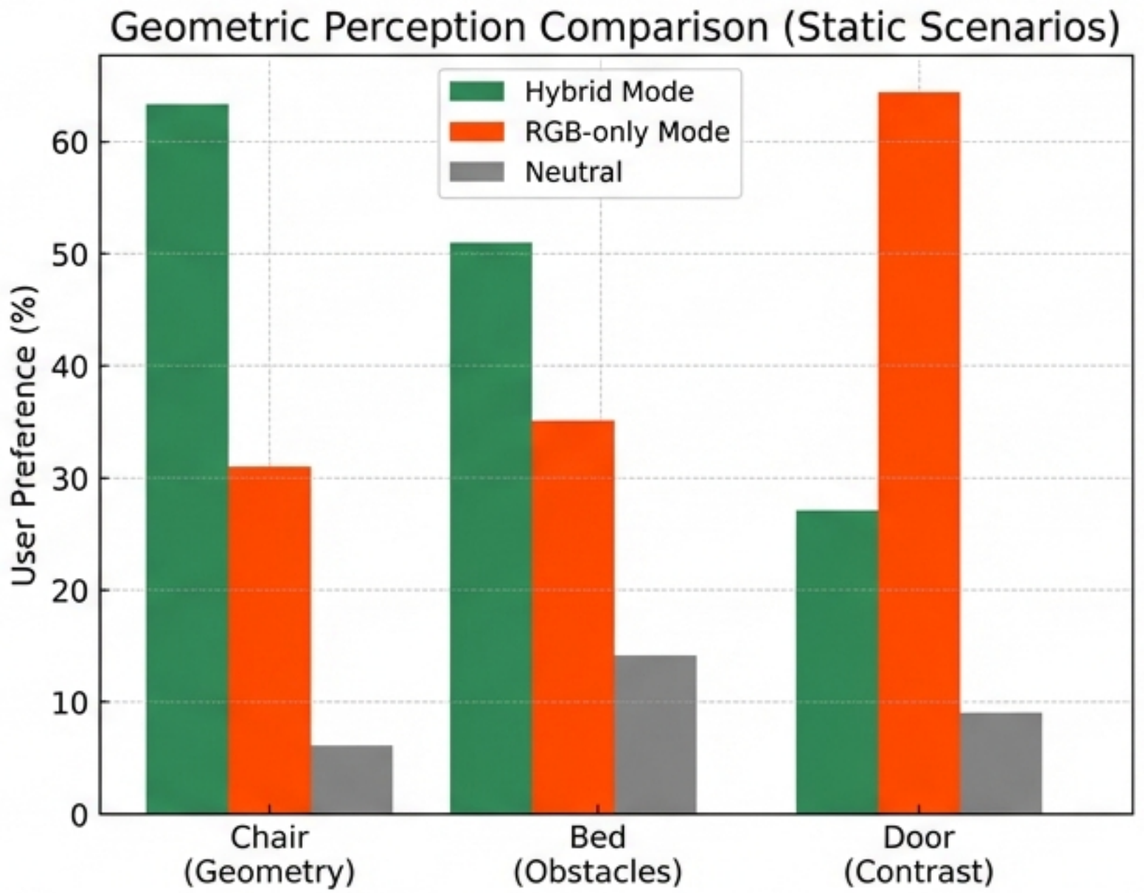
The Engineering Trade-Off:

Geometry (Chair): Hybrid Wins (63% vs 31%). Edge-enhancement defines structure.

Obstacles (Bed): Hybrid Wins (51% vs 35%). Denoising isolates scattered objects.

Contrast (Door): RGB Wins (64% vs 27%). Hybrid's aggressive filtering removes subtle texture cues.

Takeaway: Depth-Saliency excels in geometry/navigation; RGB excels in surface detail.

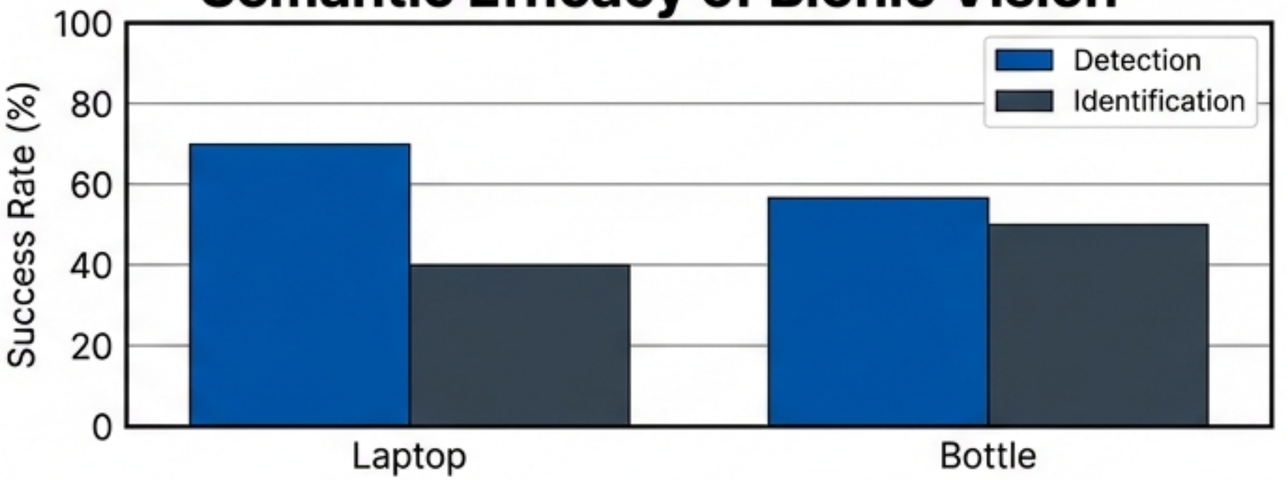


Semantic Efficacy & User Experience

Functional Vision

Detection (Noticing): >70% for Laptop. Effective Alert Mechanism.

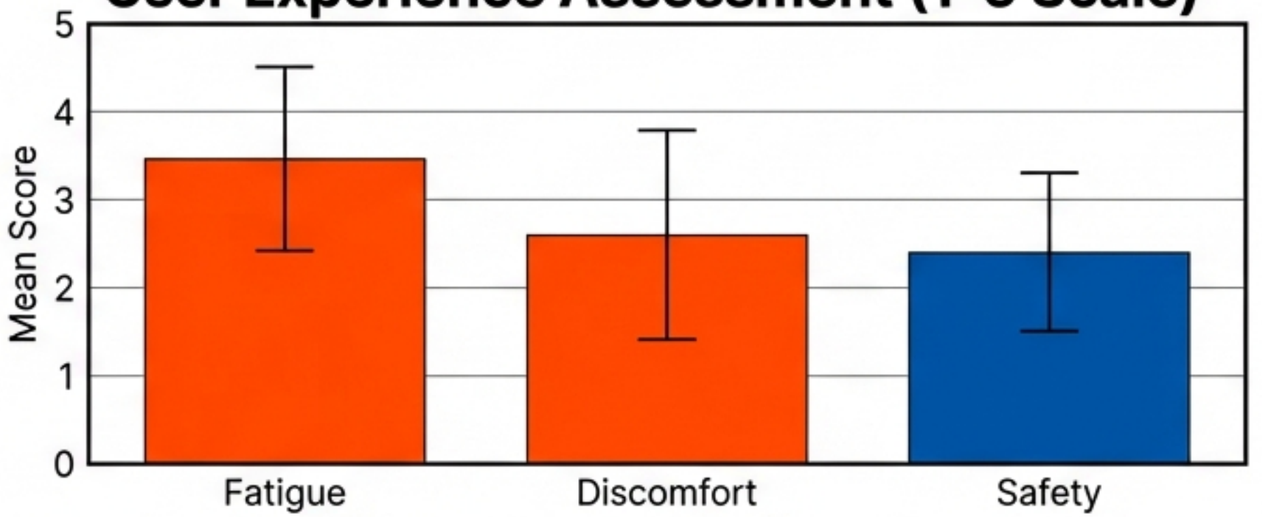
Identification (Naming): Moderate (~40-50%). Limited by 32x32 resolution.



User Experience (UX)

Mental Fatigue: 3.45/5.0. High cognitive load.

Safety Perception: 2.39/5.0. Reflects realistic user caution.



Conclusions & Future Work

Future Development

- Vision Language Models (VLM): Integration of scene description (‘A person is sitting’) via Text-to-Speech.
- Edge AI Deployment: Porting pipeline to Raspberry Pi / Jetson Nano for wearable prototypes.
- Spatial Audio: 3D sound positioning for directional cues.

Final Verdict

- The **Hybrid Depth-Saliency** approach offers the optimal trade-off between navigational safety and object recognition.
- Significantly outperforms traditional processing in dynamic environments (79.8% preference).

THANK YOU FOR YOUR KIND ATTENTION!

We hope this presentation was clear... definitely clearer than a 32x32 grid!!

Edoardo Capobianco • Simona Frascella • Michele Ivo Galella